Formula for finding square root of a complex number

$$\sqrt{a+ib}=\pm\left(\sqrt{\frac{|z|+a}{2}}+i\frac{b}{|b|}\sqrt{\frac{|z|-a}{2}}\right), \text{ where } z=a+ib \text{ and } b\neq 0.$$

Let r and θ be polar coordinates of the point $P(x,y)$ that corresponds to a non-zero complex number $z=x+iy$. The polar form or trigonometric form of a complex number P is

$$z=r(\cos\theta+i\sin\theta).$$

Properties of polar form

Property 1: If $z=r(\cos\theta+i\sin\theta)$, then $z^{-1}=\frac{1}{r}(\cos\theta-i\sin\theta)$.

Property 2: If $z_1=r_1(\cos\theta_1+i\sin\theta_1)$ and $z_2=r_2(\cos\theta_2+i\sin\theta_2)$, then $z_1z_2=r_1r_2(\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2))$.

Property3: If $z_1=r_1(\cos\theta_1+i\sin\theta_1)$ and $z_2=r_2(\cos\theta_2+i\sin\theta_2)$, then $\frac{z_1}{z_2}=\frac{r_1}{r_2}[\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)]$.

de Moivre's Theorem

(a) Given any complex number $\cos\theta+i\sin\theta$ and any integer n,

$$(\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$$

(b) If x is rational, then $\cos x\theta+i\sin x\theta$ in one of the values of $(\cos\theta+i\sin\theta)^x$

The nth roots of complex number $z=r(\cos\theta+i\sin\theta)$ are

$$z^{1/n}=r^{1/n}\left(\cos\left(\frac{\theta+2k\pi}{n}\right)+i\sin\left(\frac{\theta+2k\pi}{n}\right)\right),\ k=0,1,2,3,\ldots,n-1.$$

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